BLACK HOLE MASTER EQUATIONS

The four laws, Hawking thermodynamics, the Kerr–Newman solution, and the Teukolsky master equation.

1 FOUR LAWS OF BLACK HOLE MECHANICS

Bardeen, Carter, and Hawking (1973). These are classical identities for a stationary black hole in general relativity. They match the laws of thermodynamics if surface gravity κ is read as temperature and horizon area A as entropy. Classically the hole does not radiate, so the match is not yet thermodynamics.

Zeroth law. On the horizon of a stationary black hole, surface gravity is constant:

$$\kappa = \text{constant}. \tag{1}$$

First law. Two nearby stationary black holes, of mass M, angular momentum J, and charge Q, differ by

$$\delta M = \frac{\kappa}{8\pi}\,\delta A + \Omega_H\,\delta J + \Phi_H\,\delta Q. \tag{2}$$

Ω_H is the angular velocity of the horizon and Φ_H is the electric potential there. Restoring G puts a factor $1/G$ on the area term only: $\kappa\,\delta A/(8\pi G)$.

Second law, Hawking's area theorem. If the weak energy condition holds, the area of each event horizon never decreases. When two holes merge, the final area is at least the sum of the two initial areas:

$$\delta A \geq 0. \tag{3}$$

Third law. No finite sequence of physical operations can reduce the surface gravity to zero. Extremal holes ($\kappa = 0$) are a limit, not a state one can reach.

2 SMARR RELATION

The first law integrates, on a stationary hole, to an Euler relation among the same quantities:

$$M = 2\,\Omega_H J + \frac{\kappa A}{4\pi} + \Phi_H Q. \tag{4}$$

For Schwarzschild, $J = Q = 0$, $\kappa = 1/(4M)$, and $A = 16\pi M^2$, so (4) collapses to the identity $M = \kappa A/(4\pi)$.

3 TEMPERATURE, ENTROPY, AND THE FIRST LAW

Hawking radiation makes κ a real temperature and A a real entropy. With $\hbar$ and k_{B} restored, and still $G = c = 1$:

$$T_H = \frac{\hbar\,\kappa}{2\pi\,k_{\mathrm{B}}}, \qquad S_{\mathrm{BH}} = \frac{k_{\mathrm{B}}\,A}{4\hbar}. \tag{5}$$

In units $\hbar = k_\mathrm{B} = 1$ this is $T = \kappa/(2\pi)$ and $S = A/4$, and (2) is ordinary thermodynamics:

$$\mathrm{d}M = T\,\mathrm{d}S + \Omega_H\,\mathrm{d}J + \Phi_H\,\mathrm{d}Q. \tag{6}$$

The classical area law is then replaced by the generalized second law: matter entropy and horizon entropy together do not decrease,

$$\delta(S_\mathrm{matter} + S_\mathrm{BH}) \geq 0. \tag{7}$$

4 SCHWARZSCHILD

No spin, no charge. One parameter, the mass M.

$$r_s = 2M, \qquad A = 16\pi M^2, \qquad \kappa = \frac{1}{4M}. \tag{8}$$

The same relations in SI units:

$$r_s = \frac{2GM}{c^2}, \qquad \kappa = \frac{c^4}{4GM}. \tag{9}$$

$$T_H = \frac{\hbar\, c^3}{8\pi G M k_\mathrm{B}}. \tag{10}$$

$$S_\mathrm{BH} = \frac{k_\mathrm{B} c^3 A}{4G\hbar} = \frac{4\pi k_\mathrm{B} G M^2}{\hbar\, c}, \qquad A = \frac{16\pi G^2 M^2}{c^4}. \tag{11}$$

Temperature falls as $1/M$ and entropy grows as M^2. A solar-mass hole sits near $T_H \sim 60\,\mathrm{nK}$. Equation (5) uses the geometric κ, an inverse length. The SI temperature is (10): do not substitute the acceleration (9) into (5) without an extra factor of $1/c$.

5 EVAPORATION

Treat the hole as a blackbody of its horizon area and of temperature (10), with no gray-body correction. The power and the time to radiate the whole mass away are

$$P = \frac{\hbar c^6}{15360\,\pi G^2 M^2}, \tag{12}$$

$$t_\mathrm{evap} = \frac{5120\,\pi G^2 M^3}{\hbar c^4}. \tag{13}$$

These follow from $c^2\,\mathrm{d}M/\mathrm{d}t = -P$. Real emission is filtered by frequency-dependent gray-body factors, so the true lifetime is longer by a factor of a few and depends on which species are light compared with T_H.

6 IRREDUCIBLE MASS

Christodoulou's irreducible mass is the part of M that cannot be removed by classical extraction (the Penrose process, or superradiant scattering):

$$M^2 = M_{\mathrm{irr}}^2 + \frac{J^2}{4M_{\mathrm{irr}}^2}, \qquad A = 16\pi M_{\mathrm{irr}}^2. \tag{14}$$

The area theorem is the statement that M_{irr} never decreases. An extremal Kerr hole has $M_{\mathrm{irr}} = M/\sqrt{2}$, so at most $1 - 1/\sqrt{2} \approx 29\%$ of its mass is rotational and classically extractable.

7 KERR–NEWMAN SOLUTION

No-hair: a stationary, asymptotically flat black hole of Einstein–Maxwell theory is fixed by M, J, and Q alone. In Boyer–Lindquist coordinates, with spin parameter $a = J/M$,

$$\Delta = r^2 - 2Mr + a^2 + Q^2, \qquad \Sigma = r^2 + a^2\cos^2\theta. \tag{15}$$

The line element is

$$\begin{aligned} \mathrm{d}s^2 = &-\frac{\Delta}{\Sigma}\left(\mathrm{d}t - a\sin^2\theta\,\mathrm{d}\phi\right)^2 + \frac{\Sigma}{\Delta}\,\mathrm{d}r^2 \\ &+ \Sigma\,\mathrm{d}\theta^2 + \frac{\sin^2\theta}{\Sigma}\left[(r^2 + a^2)\mathrm{d}\phi - a\,\mathrm{d}t\right]^2. \end{aligned} \tag{16}$$

Horizons lie at $\Delta = 0$:

$$r_{\pm} = M \pm \sqrt{M^2 - a^2 - Q^2}\,. \tag{17}$$

A horizon exists only for $M^2 \geq a^2 + Q^2$. Equality is the extremal hole, $\kappa = 0$. Below that bound the singularity is naked; cosmic censorship says collapse does not produce it.

For Kerr ($Q = 0$) the horizon area, angular velocity, and surface gravity are

$$A = 4\pi\left(r_+^2 + a^2\right) = 8\pi M r_+, \tag{18}$$

$$\Omega_H = \frac{a}{2Mr_+}, \qquad \kappa = \frac{\sqrt{M^2 - a^2}}{2Mr_+}. \tag{19}$$

On a Kerr–Newman horizon the electric potential in the first law is

$$\Phi_H = \frac{Qr_+}{r_+^2 + a^2}. \tag{20}$$

8 TEUKOLSKY MASTER EQUATION

Teukolsky (1973). A single linear equation governs a massless field ψ of spin weight s on the Kerr background. It decouples and separates. The same equation covers a scalar, a neutrino, the electromagnetic field, and linearized gravity.

$$\begin{aligned}&\left[\frac{(r^2+a^2)^2}{\Delta}-a^2\sin^2\theta\right]\partial_t^2\psi+\frac{4Mar}{\Delta}\,\partial_t\partial_\phi\psi\\&+\left[\frac{a^2}{\Delta}-\frac{1}{\sin^2\theta}\right]\partial_\phi^2\psi-\Delta^{-s}\partial_r\!\left(\Delta^{s+1}\partial_r\psi\right)\\&-\frac{1}{\sin\theta}\,\partial_\theta\!\left(\sin\theta\,\partial_\theta\psi\right)-2s\left[\frac{a(r-M)}{\Delta}+\frac{i\cos\theta}{\sin^2\theta}\right]\partial_\phi\psi\\&-2s\left[\frac{M(r^2-a^2)}{\Delta}-r-ia\cos\theta\right]\partial_t\psi+(s^2\cot^2\theta-s)\psi=4\pi\,\Sigma\,T.\end{aligned}\tag{21}$$

Here $\Delta=r^2-2Mr+a^2$ is the vacuum Kerr quantity (no Q), $\Sigma=r^2+a^2\cos^2\theta$, and T is the source built from the stress-energy of the perturbation.

The spin weight selects the field:

$s=0$ scalar. $s=\pm1/2$ massless neutrino. $s=\pm1$ electromagnetic.

$s=\pm2$ linearized gravity. The field with $s=-2$ is the Weyl scalar ψ_4, which carries the outgoing gravitational-wave strain. The field with $s=+2$ is ψ_0.

Modes separate in Boyer–Lindquist coordinates:

$$\psi=e^{-i\omega t+im\phi}\,R(r)\,S(\theta).\tag{22}$$

$S(\theta)$ is a spin-weighted spheroidal harmonic. $R(r)$ obeys the radial Teukolsky equation; its complex frequencies are the quasi-normal modes of the ringdown. For $a=0$ and $s=\pm2$, (21) reduces to the Regge–Wheeler equation (odd parity) and the Zerilli equation (even parity).

A bosonic mode extracts rotational energy — it is superradiant — when

$$0<\omega<m\,\Omega_H.\tag{23}$$

NOTATION

$M,\ J,\ Q$	mass, spin, charge	$a=J/M$	Kerr parameter
$A,\ \kappa$	area, surface gravity	$\Omega_H,\ \Phi_H$	horizon angular velocity, potential
$r_s,\ r_\pm$	Schwarzschild and horizon radii	$T_H,\ S_{\mathrm{BH}}$	temperature and entropy
M_{irr}	irreducible mass	$\psi,\ s$	master field, spin weight
$\omega,\ m$	frequency, azimuthal number		

SOURCES

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